भारतीय सूचना प्रौद्योगिकी संस्थान गुवाहाटी

INDIAN INSTITUTE OF INFORMATION TECHNOLOGY GUWAHATI

CS 360: Machine Learning Lab

Practice Assignment 5

Instructions: This is only for practice. Complete it by 12:00 PM today. Your completion will be reviewed by the Teaching Assistants.

1. Considering the **Diabetes prediction** dataset for classification of patients to have diabetes or not, write a program to do the followings:
   (a) Randomly split the dataset into a 80:20 training:testing splits, and consider 10% of the training set as validation set.
   (b) Design the hypothesis ($\hat{y}$) (both with and without in-built packages) to classify diabetic and non-diabetic patients using Logistic regression model to estimation of the model parameters (θ) or (ω), by applying the followings:
   - Batch gradient descent method.
   - classification threshold=0.5 .

   without using in-built Python packages /libraries for both.
   - Consider model parameters are to be initialized (θ) or (ω) to zero.
   - Apply hyper-parameter tuning for selecting the best model for the following hyper-parameters, considering the given values:
     1. learning rate, α: $\{0.0001, 0.1\}$
     2. ρ: $\{.0.001, 0.01\}$ where, ρ signifies absolute error-difference in two consecutive epochs.
     3. epoch , (T): $\{50,100\}$

   (c) Calculate and visualize the confusion matrix, considering the designed hypothesis (($\hat{y}$) and the actual targets (y). From the confusion matrix obtained, calculate and report the overall accuracy, class-wise accuracy, precision and recall values.
   (d) Obtain hypotheses for varying thresholds of 0.3 and 0.7, and generate the confusion matrices respective to each hypothesis.
   (e) Compute and compare the precision and recall values w.r.t. all three threshold values. Write your observations on how the changing thresholds effect the false-positive (FP) and false-negative (FN) values.
   (f) Repeat 1(a)- 1(b) by applying K-fold cross validation for $K = 5$. Consider 10% of the training set as validation set.
   (g) Calculate and report confusion matrices for all 5-folds, and the overall accuracy, class-wise accuracy, precision and recall values, with respect to average of the k-folds and the best fold.
   (h) Report your observations on whether the random training:testing splitting and K-fold cross validation has distinct effect on the model performance (in terms of overall accuracy, class-wise accuracy, precision and recall-values).